

Bill of Materials

ASHA Sahayak — offline Hindi voice assistant, prototype unit

Build of record · 20 August 2026
Quantities are per one assembled unit
Costs in ₹ INR, inclusive of 18% GST

The prototype contains **no custom hardware**. Every part is an off-the-shelf module bought from a distributor and joined with jumper leads — a deliberate choice, so the engineering effort went into the software stack rather than into a board spin we would have had to redo once the software settled. The custom items are listed separately as the production step.

01 Prototype unit — as built and measured

ITEM	PART NUMBER	QTY	UNIT COST	TYPE	FUNCTION IN THIS BUILD
Jetson Orin Nano Super Developer Kit, 8 GB	945-13766-0000-000	1	37,712	OTS	Compute. Bundles the Orin Nano 8 GB module (P3767), the reference carrier board (699-13768) and a 19 V / 45 W supply. Runs every model on the device; 8 GB unified memory is the binding constraint the software was built around.
microSDXC card, 128 GB UHS-I	SanDisk SK128 · manfid 0x03	1	1,200	OTS	Boot and root filesystem. Holds JetPack 6.2.1 plus 5.2 GB of model weights. Identifiers are as reported by the installed card; any A2-rated 128 GB card substitutes, and the A2 rating is what keeps model load time at boot reasonable.
Arducam 8 MP autofocus USB camera with microphone	B0197 · succ. B0447	1	4,800	OTS	Microphone today, sensor tomorrow. The shipping build is voice-only, so the IMX179 sensor is idle — but it is the sensor the planned screening trial (neonatal jaundice first, under MoH / ICMR protocol) depends on. The capability is already paid for in this build; only the approval is outstanding.
USB speaker, UAC 1.0 class-compliant	Jieli UACDemoV1.0 · 4c4a:4155	1	700	OTS	Answer playback at 22 kHz. Class-compliant, so no driver and no vendor lock — any UAC device substitutes.
Powered USB 3.0 hub, 4-port	Realtek RTS5411 · 0bda:0489	1	600	OTS	Fans out the carrier's USB to camera, speaker and service port without browning out the audio device under load.
2.8" SPI TFT display, 320×240, with resistive touch	ILI9341 + XPT2046	1	800	OTS	Shows the question and answer in Devanagari. Driven over SPI from the 40-pin header; text is shaped with HarfBuzz because bitmap fonts cannot render Hindi conjuncts.
Momentary push button, 16 mm panel mount, NO	generic 16 mm NO	1	120	OTS	The single control. Wired to header pin 7 (GP167): one press starts recording, the next ends it, and 3.5 s of silence ends it anyway if she forgets. The design does not depend on the pin reporting a sustained press, so it works with any momentary contact.
Jumper leads, female–female 20 cm	generic Dupont 2.54 mm	1 set	100	OTS	Header wiring for the button and the display. Replaced by the custom carrier in production.
Enclosure, ABS project box	generic 158×90×60 mm	1	450	OTS	Holds the assembly, the button and the display face. Not field-rated — a moulded, sealed case is a production item.
Prototype unit total — 9 line items, all off-the-shelf			46,482		Single-unit distributor pricing, India, inclusive of 18% GST.

Why nothing here is custom yet

A custom carrier board designed before the software was finished would have been wrong twice: the voice build does not need the camera for capture, and the trigger pin turned out to need input conditioning the reference carrier does not provide — it cannot hold a sustained press, which is why recording is a toggle rather than hold-to-talk. Both were found by running the real workload on off-the-shelf parts. The production board below is specified **because** those answers are now known — it fixes the pull-down in the pinmux, integrates an audio path that is currently three separate USB devices, and **keeps a camera interface** so the screening SKU is a populate option rather than a second board.

02 Production unit — target at 10,000 units

ITEM	PART NUMBER	QTY	UNIT COST	TYPE	CHANGE FROM PROTOTYPE
8 GB ARM system-on-module	P3767-0005 or equivalent	1	7,000	OTS	Module only, no developer carrier. The stack needs 4.2 GB and 5.1 W and has no CUDA-only dependency, so a commodity 8 GB ARM SoM is a port, not a rewrite — which is what takes this line from ₹37,712 to ₹7,000.
Carrier board, 4-layer	AS-CB-01 (to design)	1	1,400	CUSTOM	SoM connector, codec, button with a proper pull-down, display FPC, power. Removes the hub and all jumper wiring.
Enclosure, injection moulded, IP54	AS-EN-01 (to tool)	1	700	CUSTOM	Field-rated case with speaker grille and sealed button. Tooling is a one-time cost amortised across the run, not in this unit price.
MEMS microphone board	generic I2S MEMS	1	180	OTS	Takes over audio capture from the camera module at 4% of its cost. The camera itself becomes an option rather than a requirement — see the vision SKU note below.
Speaker driver, 2 W 8 Ω, with amplifier	generic + PAM8302	1	220	OTS	On-board audio out; retires the USB speaker and one hub port.
2.8" SPI TFT display, 320×240	ILI9341, FPC mount	1	600	OTS	Same panel, ribbon-mounted to the carrier instead of jumpered.
eMMC storage, 64 GB	on-module eMMC	1	750	OTS	Replaces the microSD card. 5.2 GB of weights plus the OS fits comfortably; removes the commonest field failure on SD-booted devices.
Battery, 3.7 V 10,000 mAh, with 2 A charge controller	generic Li-ion + 2 A charger IC	1	900	OTS	New in production; the prototype has no battery and runs from the barrel jack. 37 Wh gives about 6 hours of continuous querying at 5.1 W, and a full shift at a realistic duty cycle, since the board idles at 3.4 W. A 2 A charger is specified so it refills overnight — a 1 A part would need more than ten hours.
Assembly, test and pack	—	1	250	CUSTOM	Includes flashing the image, loading the knowledge base and running the fixed ten-question accuracy check on every unit before it ships.
Production unit total — 3 custom items, 6 off-the-shelf			12,000		Target at 10,000 units; excludes one-time tooling and NRE.

03 Notes

- **No radio is used.** The developer kit's M.2 Wi-Fi/Bluetooth card is present and enumerates, but the application never opens a socket and the device is airplane-mode capable by construction. The production carrier omits the M.2 slot entirely.
- **Vision SKU.** The production carrier retains a camera interface. Populating it with an 8 MP autofocus module adds roughly ₹900 at volume. It is deliberately **not** in the ₹12,000 base: no camera ships until a screening aid has cleared Ministry of Health and ICMR validation, and the device presents any such result as a reason to refer, never as a diagnosis.
- **Prices** are single-unit Indian distributor prices as of August 2026, inclusive of 18% GST; the Jetson kit is listed at ₹31,959 ex-GST. Production figures are targets based on component pricing at 10,000 units, not quotations.
- **Nothing in the software is tied to this hardware.** Speech, translation and synthesis run through ONNX Runtime and CTranslate2, and the language model through llama.cpp — all with working CPU paths. That is what makes the SoM line above substitutable, and it is the single largest lever on unit cost.